

Long-Range Dependence and Self-Similarity

This modern and comprehensive guide to long-range dependence and self-similarity starts with rigorous coverage of the basics, then moves on to cover more specialized, up-to-date topics central to current research. These topics concern, but are not limited to, physical models that give rise to long-range dependence and self-similarity; central and non-central limit theorems for long-range dependent series, and the limiting Hermite processes; fractional Brownian motion and its stochastic calculus; several celebrated decompositions of fractional Brownian motion; multidimensional models for long-range dependence and self-similarity; and maximum likelihood estimation methods for long-range dependent time series. Designed for graduate students and researchers, each chapter of the book is supplemented by numerous exercises, some designed to test the reader's understanding, while others invite the reader to consider some of the open research problems in the field today.

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CAMBRIDGE
UNIVERSITY PRESS

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University Printing House, Cambridge CB2 8BS, United Kingdom

One Liberty Plaza, 20th Floor, New York, NY 10006, USA

477 Williamstown Road, Port Melbourne, VIC 3207, Australia

4843/24, 2nd Floor, Ansari Road, Daryaganj, Delhi – 110002, India

79 Anson Road, #06–04/06, Singapore 079906

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www.cambridge.org

Information on this title: www.cambridge.org/9781107039469

DOI: 10.1017/9781139600347

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First published 2017

Printed in the United States of America by Sheridan Books, Inc.

A catalog record for this publication is available from the British Library.

ISBN 978-1-107-03946-9 Hardback

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